Data Analyst Internship

A 6-Week Professional Internship Presentation Report

INTERNSHIP DURATION
17 July – 28 August 2026 (6 Weeks)

STUDENT CREDENTIALS

Student Name: Adamya Dixit
Enrollment / Roll No: 25SCS1003003698
Degree & Batch: B.Tech (CSE) | 2026–27
University: IILM University, Greater Noida, U.P.

HOST ORGANIZATION

Host Partner: BeeSkilled
Accreditation: Registered with AICTE & MSME
Program Head: Ms. Simranjeet Kaur (Founder)

Internship Overview & Organization Profile

6-WEEK DATA ANALYST INTERNSHIP

• Structured, project-based virtual learning designed to apply academic CSE knowledge to commercial data pipelines.

• Role: Data Analyst Trainee — responsible for compiling data, handling quality anomalies, creating computed features, and building reports.

• Timeline & Effort: Full-time curriculum across July and August 2026, comprising weekly technical exercises and an end-to-end business analytics capstone project.

• Analytical Focus Areas: Multi-source business reporting, database wrangling with Python, and calculated DAX visual modeling in Power BI.

HOST PROFILE: BEESKILLED

• Vision: Online skill-development and hands-on internship platform operating under the tagline 'Tech and Internships'.

• Official Registry: Fully registered and certified under the AICTE National Internship Portal and the Ministry of Micro, Small and Medium Enterprises (MSME).

• Curriculum Delivery: Directed by Ms. Simranjeet Kaur (Founder & Program Head) using project-based coursework to prepare students for core technical careers.

• Strategic Aim: Bridging the theoretical academic foundations of Computer Science with real-world enterprise dataset challenges.

Problem Statement, Objectives & Scope

PROBLEM STATEMENT

THE INDUSTRY CHALLENGE

- Organizations generate massive volumes of raw operational and transactional data every day.
- Raw datasets are highly prone to formatting discrepancies, duplicates, and missing records.
- This operational data carries zero business utility until it is systematically cleaned, structured, analyzed, and visualized.

KEY FOCUS OUTCOMES

CORE INTERNSHIP OBJECTIVES

- Establish a rigorous analytical workflow moving from raw collection to business insights.
- Master spreadsheet aggregation, conditional formulas, and pivot-table dashboards.
- Acquire hands-on proficiency in Python and Pandas for dataset wrangling and statistics.
- Develop calculated measures and interactive report views using Power BI and DAX.

SCOPE OF ANALYSIS

PROJECT BOUNDARIES

- Analysis of structured retail, fitness-tracking, and financial transactional datasets.
- Focused on modular, off-line analytical pipelines (Excel, Pandas, Power BI).
- Did not involve proprietary live production databases, but simulated real-world scenarios.
- Emphasized translating data findings into executive strategic suggestions.

Technologies & Structured Analytics Workflow

Tool	Purpose	Application during Internship
Microsoft Excel	Spreadsheet-based data organization and analysis	Built pivot-table dashboard analyzing total revenue, sales by category, yearly trends, and segment-wise revenue from a 51,290-record orders dataset.
Python	General-purpose programming for data analysis tasks	Used for dataset loading, inspection, and data-cleaning scripts across Weeks 2 and 3.
Pandas	Data manipulation — reading, inspecting, cleaning, and transforming data	Inspected a 200-row sales dataset using head()/info()/describe(); cleaned a 32-row exercise dataset by handling missing values, duplicates, and date formatting; created a derived column and filtered records.
Power BI	Business intelligence dashboarding and calculated (DAX) measures	Built DAX measures — Total Sales, Total Profit, Total Units Sold, Profit Margin, Average Order Value, and Loss-Making Orders — for the Week 4 capstone dashboard.

THE END-TO-END DATA ANALYTICS WORKFLOW

1. Raw Source → 2. Data Ingestion → 3. Cleaning → 4. Wrangling → 5. Analysis → 6. Visualization → 7. Strategy

Week 1: Excel Sales Performance Dashboard

High-Volume Transaction Summarization on Global Superstore Dataset (51,290 Records, 24 Columns)

TECHNOLOGY SALES
$4.74 Million
Segment Leader by Revenue

FURNITURE SALES
$4.11 Million
Strong Secondary Contributor

OFFICE SUPPLIES
$3.79 Million
High Volume, Medium Ticket

Customer Segment Performance

Customer Segment	Pivot Total Sales
Consumer Segment	**$6.51 Million**
Corporate Segment	**$3.82 Million**
Home Office Segment	**$2.31 Million**

Yearly Revenue Trend Analysis

Fiscal Year	Aggregated Revenue
Calendar Year 2012	**$2.26 Million**
Calendar Year 2013	**$2.68 Million**
Calendar Year 2014	**$3.41 Million**
Calendar Year 2015	**$4.30 Million (Steady Growth)**

Week 2: Python & Pandas Dataset Exploration

PROGRAMMATIC INSPECTION PIPELINE

• Injected a 200-row, 10-column retail transaction dataset into Pandas.

• Variables Evaluated: order_id, customer_name, order_date, category, sub_category, product_name, quantity, unit_price, total_price, and region.

• Primary inspection methods used — head(), info(), shape, and columns — to audit structural validity, identify empty fields, and verify data schema.

• Successfully confirmed clean column data types, uniform formatting, and zero missing cells in the structural verification phase.

DESCRIPTIVE STATISTICS SUMMARY (df.descri

MEAN QUANTITY PER ORDER

4.91 Units

Standard transaction basket size

MEAN UNIT TRADING PRICE

$2,421.50

High-value enterprise items

MEAN TOTAL TRANSACTION VALUE

$12,101.54

Healthy average basket size

Maximum Single Order Value recorded in the dataset: $47,940

Week 3: Programmatic Data Wrangling & Cleaning

CASE STUDY: RAW WORKOUT DATASET

• Raw Dataset Profile: 32 rows, 5 columns (Duration, Date, Pulse, Maxpulse, Calories).

• Quality Audit Findings: Uncovered 1 critical duplicate record, 1 blank row date cell, and 2 empty calorie readings.

Cleaning Procedures Implemented:
- Duplicate Removal: Dropped identical entry.
- Missing Date Imputation: Resolved blank cell via linear interpolation.
- Missing Calories Resolution: Filled empty cells with column mean.

• Schema Standardization: Converted Date field from inconsistent plain text format to pandas proper datetime series.

TRANSFORMATION & EXPORT PIPELINE

Metric	Before	After
Total Records	32 rows	31 rows (1 dropped)
Missing Values	3 empty cells	0 (Fully resolved)
Calculated Columns	None	Calories_Per_Minute

Feature Engineering & Subset Filtering

1. Engineered calculated field 'Calories_Per_Minute' = Calories / Duration to evaluate exercise intensity.

2. Exported pristine clean dataset as 'cleaned_data.csv' (31 rows, 6 columns).

3. Filtered workout records to isolate high-intensity exercises (>300 Calories) and exported separate subset as 'filtered_data.csv'.

Week 4: End-to-End Business Analytics Capstone

CAPSTONE DATASET PROFILE

• Dataset Volume & Dimensions: 700 structured records across 16 commercial tracking columns.

• Primary Variables: Segment (Consumer/Enterprise/Gov/etc), Country, Product, Discount Band, Units Sold, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month, Year.

• Practical Domain: Financial sales audit simulating enterprise operations across multiple products and global territories.

• Analytical Goal: Build a fully functional Power BI dashboard with calculated DAX KPIs and extract high-level strategic recommendations.

DATA CLEANING & INTEGRITY AUDITING

HANDLING MISSING DISCOUNT BANDS

Identified 53 records with blank Discount Bands. Investigation revealed these had exactly 0 discount values. Re-classified them as 'No Discount' rather than dropping, successfully preserving 100% of transaction data.

NUMERIC INTEGRITY AUDITING

Ran cross-column mathematical verification across all 700 rows:

data integrity.

Helper Column Created: Year-Month for monthly trends. Exported: Week_4_Cleaned_Data.csv

Capstone: Power BI Calculated DAX Measures

Summary of Business performance indicators formulated using computed DAX expressions in Power BI

TOTAL SALES REVENUE

$118,726,350

Aggregate sales value across segments

TOTAL NET PROFIT

$16,893,702

Consolidated net margin after COGS

TOTAL UNITS SOLD

1,125,806 Units

Consolidated product sales volume

AVERAGE ORDER VALUE

$169,609

Mean transaction deal size

HEALTHY PROFIT MARGIN

14.2%

Average percentage margin on sales

TOTAL COMPLETED ORDERS

700 Transactions

Full transaction audit count

Capstone: Key Business Findings & Data Insights

CRITICAL BUSINESS DISCOVERIES

• Government Segment Dominance: Government was the absolute strongest division, contributing $11.39M to profits.

• Enterprise Margin Drainage: Enterprise emerged as the only segment that recorded an overall loss of -$614,546, suggesting severe pricing or cost issues.

• Top Geographical Market: The USA generated the highest regional revenue, topping all global countries at $25.03M.

• Product Performance Divergence: Product 'Paseo' was the best financial asset in sales and margins, while 'Carretera' was the weakest performer in both.

• Loss-making Order Correlates: Identified that 58 of 700 orders (8.3%) lost money, with strong ties to Medium/High discount bands.

• Seasonal Demand Peaks: Visual analysis revealed major calendar demand surges in October and December 2014, and a severe slump in September 2013.

SEGMENT PERFORMANCE SUMMARY

Segment Profile	Net Profit/Loss	Trading Status
Government	**$11,390,000**	**Top Performer**
Enterprise	**-$614,546**	**Critical Loss**
Others (Combined)	**$6,118,248**	**Profitable**

ANALYST INSIGHT: DISCOUNT THRESHOLDS

Exploratory logic audits confirmed that transactions under 'Medium' and 'High' discount tiers represent 100% of loss-making orders (58 total). This highlights the need to re-evaluate bulk concession rules.

Recommendations & Internship Outcomes

STRATEGIC RECOMMENDATIONS

• Discount Margin Cap: Introduce strict margin thresholds for bulk deals to curb deficit orders (8.3% of total) in High/Medium discount bands.

• Enterprise Audit: Conduct an operational review of the Enterprise division's cost structure to address the overall loss of -$614,546.

• Standardized Carretera Pricing: Overhaul 'Carretera's pricing structure, as it showed negative trends in both sales and profits.

• Seasonal Resource Alignment: Align warehouse and supply chain resources with the observed peak seasons of October and December.

• Post-Mortem Analysis on September 2013: Investigate the severe performance drop in September 2013 to protect against similar market declines.

TECHNICAL LEARNING OUTCOMES

• Spreadsheet Intelligence: Mastered Excel pivot tables, dynamic aggregation, and conditional dashboard formatting.

• Programmatic Wrangling: Mastered Python, Pandas, and numpy workflows to drop duplicates, interpolate gaps, and engineer features.

• Relational BI Modeling: Built dynamic DAX calculated measures and interactive report filters using Power BI.

• Systematic Data Auditing: Implemented strict multi-column logic checks to ensure 100% database mathematical integrity.

• Business Storytelling: Learned to bridge mathematical outputs with high-level corporate strategy suggestions.

Official Credentials & Internship Work Proof

Verified credentials confirming completion of the 6-Week Data Analyst program

INTERNSHIP OFFER LETTER

Date: 17/07/2026

To,
Adamya Dixit

Subject: Offer For Internship

Dear Adamya Dixit ,

We are pleased to offer you an online internship with **Beeskilled** in the domain of Data Analyst. The internship will commence from 17/07/2026 and end on 28/08/2026.

During this period, you will be required to work on assigned projects and complete the tasks within deadlines. This internship is aimed at enhancing your practical knowledge and hands-on experience.

We look forward to your positive response and hope this internship will contribute meaningfully to your career growth.

Best regards,

Skaur

Simranjeet Kaur
Founder & Program Head

AICTE & MSME REG.

VERIFICATION METRICS

Student Name:
Adamya Dixit

Profile:
Data Analyst

Provider:
BeeSkilled

Duration:
6 Weeks

Issue Date:
28 August 2026

Offer Ref:
17 July 2026

Reg Status:
AICTE & MSME Approved

THANK YOU!

Data Analyst Professional Internship Presentation

Presenter: Adamya Dixit | Roll No: 25SCS1003003698
Degree: B.Tech (Computer Science & Engineering) | Batch: 2026-27
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

Academic Institution: IILM University | Industry Partner: BeeSkilled